Terrain-Dependent Effects of a Nomadic Aerial 5G Base Station on Ground-Robot Command-and-Control: A Hardware-in-the-Loop Emulation Study

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***Abstract*—Aerial 5G base stations (gNBs) carried by unmanned aerial vehicles (UAVs) are a promising way to deliver command-and-control (C2) and sensor backhaul to ground robots in search-and-rescue (SAR) missions where fixed infrastructure is absent. A natural question is whether the gNB should *hover* over the search area or *roam* with the robots. We study this using a hardware-in-the-loop (HIL) emulation that couples a real 5G stand-alone stack (OpenAirInterface gNB and user equipment over the RF simulator, Open5GS core) to a physics simulator (Gazebo) through an air-to-ground (A2G) channel bridge that applies per-link path loss, terrain-specific non-line-of-sight (NLoS) excess, and log-normal shadowing in real time; per-robot network namespaces force every C2 packet across the emulated air interface. Across four terrains (open field, urban, forest, hills) each robot runs a 6-hour trial split into a 3-hour *static*-gNB phase and a 3-hour *nomadic*-gNB phase (continuous 3-D random walk). We find the effect of gNB mobility on C2 availability is strongly *terrain-dependent*: nomadism helps a weak-link robot in line-of-sight (open field, $+32$ percentage points), is neutral under localized building NLoS (urban), and is progressively harmful as blockage becomes more pervasive (hills -10.6 pp, forest -22 pp). We further show that maintaining C2 in NLoS at all requires a coverage controller that reacts to *measured* path loss rather than geometric distance, and that deep outdoor-to-indoor penetration is effectively uncoverable at this link budget. We release the emulation stack, the experiment automation, and the dataset.**

***Index Terms*—UAV communications, 5G NR, aerial base station, air-to-ground channel, search and rescue, hardware-in-the-loop emulation, OpenAirInterface, robot command and control.**

I. Introduction

Search-and-rescue (SAR) operations increasingly deploy teams of ground robots to explore hazardous areas, streaming video and sensor data back to operators while receiving command-and-control (C2) updates such as waypoints and emergency stops. In collapsed, remote, or disaster-struck environments, terrestrial cellular infrastructure is frequently unavailable, motivating *aerial* base stations: a 5G New Radio (NR) gNB carried by an unmanned aerial vehicle (UAV) that provides a nomadic radio cell over the search area [1], [2]. Because the gNB is airborne and mobile, a mission planner faces a placement decision that a fixed tower never poses: should the aerial gNB *hover* at a central vantage point, or *roam* to follow the robots?

The answer is not obvious. A higher or repositioning aerial gNB can shorten the air-to-ground (A2G) link and improve the line-of-sight (LoS) probability, which argues for mobility [3]. But in cluttered environments the same mobility can drive the gNB *behind* obstacles relative to a robot, or lengthen the propagation path through foliage, degrading the link. The trade-off depends on the geometry *and* on the physical mechanism of blockage – buildings, trees, terrain, or building walls – each of which loads the channel differently.

Answering this question with fidelity is hard. Pure network simulators abstract the protocol stack; pure robotics simulators abstract the radio. We instead build a *hardware-in-the-loop* (HIL) emulation that runs an unmodified 5G stand-alone (SA) stack – an OpenAirInterface (OAI) gNB and two user equipments (UEs) over the RF simulator, with an Open5GS core – and couples it to the Gazebo physics simulator through a real-time A2G channel bridge. The bridge reads the 3-D poses of the UAV-gNB and the robots from the simulator and, every 0.5 s, sets each link's attenuation in the RF simulator from an A2G path-loss model plus a terrain-specific NLoS term and correlated shadow fading. Crucially, we place each UE's tunnel interface in its own Linux network namespace so that C2 traffic (base $\rightarrow$ robot) genuinely traverses the emulated air interface in both directions rather than short-circuiting over the host loopback. The result is an honest, protocol-complete measurement of C2 availability, throughput, latency, and radio-link stability as a function of gNB and robot motion in a chosen terrain.

Contributions.

- A reproducible HIL emulation coupling a full OAI/Open5GS 5G SA stack to Gazebo through a real-time A2G channel bridge, with namespace isolation that makes the downlink C2 path physically honest (Section III).
- A terrain-parameterized A2G channel model that shares a single obstacle geometry between the physics world and the radio channel for urban buildings, forest foliage, knife-edge terrain, and 3GPP outdoor-to-indoor (O2I)

penetration (Section IV).

- An NLoS-aware coverage controller that keeps robots connected by reacting to *measured* path loss, without which cluttered terrains suffer an unrecoverable radio-link-failure (RLF) cascade (Section V).
- A controlled static-vs-nomadic gNB study over four terrains and 6 hours each, establishing that the sign of the mobility effect flips with the pervasiveness of blockage, and characterizing the O2I coverage limit (Section VII).

II. Related Work

UAV-assisted communications. The use of UAVs as flying base stations or relays is surveyed in [1], [2]. A central result is the existence of an optimal altitude that balances free-space loss against LoS probability for a low-altitude platform [3]. Most of this literature optimizes *placement* analytically or in system-level simulation under statistical channel models; comparatively little work evaluates a *running* 5G stack driving *physical* robot mobility, which is our focus.

A2G channel models. The elevation-angle-dependent LoS probability and excess-loss formulation of [3] and the 3GPP channel models [4], [5] underpin our path-loss core. For terrain-specific excess we use the Weissberger vegetation model [6] for foliage and knife-edge diffraction [7] for hills, and the 3GPP O2I penetration term [4] for indoor.

Emulation platforms. OpenAirInterface provides an open, standard-compliant 5G stack with an RF simulator that replaces radios with a TCP channel, enabling software-only experiments [8]. Open5GS supplies the 5G core. Namespace- and container-based network emulation in the style of Mininet [9] lets us isolate each UE's data path. Gazebo provides rigid-body physics and sensor simulation for the robots [10]. Our contribution is not any one component but their real-time coupling through a geometry-consistent A2G channel, and the namespace design that makes the C2 downlink honest.

Robot connectivity and SAR. Connectivity-aware planning keeps robots within coverage of a base or relay. We instead ask the dual question – how a mobile base should behave given roaming robots – and measure it end-to-end on a real stack rather than under an abstract link model.

III. System Architecture

Figure 1 shows the emulation. Three subsystems run concurrently on one host and are joined by the channel bridge.

5G stack. An OAI gNB and two OAI UEs run over the RF simulator on 5G NR band n78 (3.32 GHz, 40 MHz, numerology 1), attached to an Open5GS 5G core. The RF simulator exposes a telnet control interface through which the per-link channel attenuation can be set at run time.

Physics. Gazebo simulates one UAV (the aerial gNB) and two differential-drive ground robots, publishing all 3-D poses on a pose topic. Robots are driven by velocity commands with pose feedback, so motion is kinematic rather than teleported.

Channel bridge. A Python process subscribes to the pose topic and, at 2 Hz, computes each link's attenuation (Section IV) and writes it to the RF simulator over telnet as a positive path-loss magnitude. The same obstacle geometry that Gazebo renders is imported by the bridge, so the radio channel and the visible world are consistent by construction.

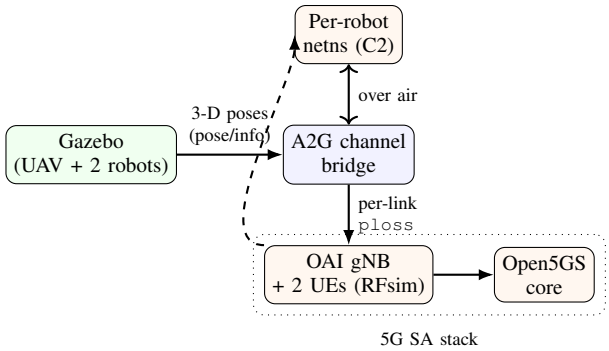


Fig. 1. HIL emulation. Gazebo streams UAV/robot poses to the channel bridge, which sets each RF-simulator link's attenuation in real time. UE tunnel interfaces live in per-robot network namespaces so the C2 downlink crosses the air interface honestly.

Namespace isolation. Without isolation, both UE IP addresses are local to the host, so downlink traffic to a robot (C2 commands, TCP ACKs) is delivered over loopback and never touches the radio. We move each UE's tunnel device into its own network namespace; the UE process keeps its open file descriptor across the move, so the stack is untouched but base-to-robot traffic now genuinely traverses core, gNB, and the emulated air interface. This makes the C2 metric an honest measurement.

IV. Air-to-Ground Channel Model

For a robot at horizontal distance d from the UAV nadir and UAV height h, the 3-D distance is $r = \sqrt{d^2 + (h - h_u)^2}$ with UE height h_u. The core path loss combines free-space loss with an elevation-angle-dependent LoS/NLoS mixture [3]:

$$\mathrm{FSPL} = 20\log_{10} r + 20\log_{10} f - 147.55, \tag{1}$$

$$p_{\mathrm{LoS}} = \frac{1}{1 + a\, e^{-b(\theta - a)}}, \quad \theta = \arctan\frac{h - h_u}{d}, \tag{2}$$

$$\mathrm{PL} = \mathrm{FSPL} + p_{\mathrm{LoS}}\eta_{\mathrm{LoS}} + (1 - p_{\mathrm{LoS}})\eta_{\mathrm{NLoS}}, \tag{3}$$

where f is the carrier frequency and $(a, b, \eta_{\mathrm{LoS}}, \eta_{\mathrm{NLoS}})$ are environment constants. The bridge reports the delta of PL relative to a fixed reference distance, adds a per-link baseline that calibrates the RF-simulator operating point, and clamps the total to the model range.

Terrain-specific NLoS excess. Each world contributes an additive term Δ_{env} evaluated on the *shared* geometry:

- *Urban:* building footprints; if the UAV→robot segment intersects a building, a hard NLoS excess applies (LoS is otherwise clear).
- *Forest:* Δ_{env} grows with the foliage path length through tree canopies (Weissberger form [6]).
- *Hills:* knife-edge diffraction loss over terrain ridges [7].
- *Indoor:* a 3GPP O2I penetration term (fixed wall loss plus a per-metre depth loss) whenever the robot is inside the building footprint [4].

Shadow fading. On top of the smoothed mean we add correlated log-normal shadowing (larger variance under NLoS), decorrelated over distance moved so that a stationary robot holds a steady shadow offset and only motion mixes in new fading.

V. NLoS-Aware Coverage Controller

A naive coverage rule keeps each robot within a geometric *safe radius* computed from the open-field model at the current UAV altitude. We found this fails badly in every NLoS terrain: a robot can sit well inside the safe radius yet be blocked (path loss near the RLF cliff), because the distance rule is blind to buildings, trees, and terrain. When such a robot loses the link, an RLF is declared; and because a UE cannot complete cell (re)synchronization while buried in deep NLoS, the recovery – restarting the UE softmodem – *fails to re-attach* ("tunnel never reappeared"). The controller then retries indefinitely, and the robot stays dark. In a first full run this produced 86 failed re-attachments in the urban world and near-zero availability for the weak robot.

We therefore make the controller react to the *measured* path loss p (read cheaply from the bridge log tail at driver rate), not to distance:

- *Soft:* if p is elevated, steer the robot *toward* the gNB ground track (raising the elevation angle and reducing NLoS depth), with reactive obstacle avoidance still routing around walls.
- *Hard:* if p stays near the cliff, teleport the robot to a small radius from the gNB (a steep elevation angle gives near-certain LoS in every world) *before* it dies, since a re-attach cannot rescue a UE with no signal.

For the indoor mission, where the robot *must* occupy the building, pulling it toward the gNB would defeat the task; instead a robot whose p nears the cliff *retreats along its reversed entry path toward the door*, where O2I loss falls and the UE survives, then re-enters. This prevention-first design converts the unrecoverable cascade into rare, short, recoverable outages (Section VII).

VI. Experimental Methodology

Worlds. Open field (LoS baseline), urban (building NLoS), forest (foliage), hills (diffraction), and indoor (O2I). All share the identical robot, UAV, and radio configuration; only the world geometry and its channel term differ.

Protocol. Each world runs a single 6-hour trial in two phases. In *phase 1* (0–3 h) the gNB hovers statically. In *phase 2* (3–6 h) the gNB performs a continuous 3-D random walk over a bounded box (never dwelling). Both ground robots free-patrol throughout, so the only variable between phases is gNB mobility. Figure 2 contrasts the two regimes.

Metrics. Every ∼15–60 s we log, per robot: C2 round-trip loss and latency (ICMP base→robot over the air), served TCP and fixed-rate UDP uplink throughput (modelling a sensor/video feed), UDP loss and jitter, and the modem's SS-RSRP, SS-SINR, and downlink BLER, together with the current path loss and the gNB–robot slant distance. We define

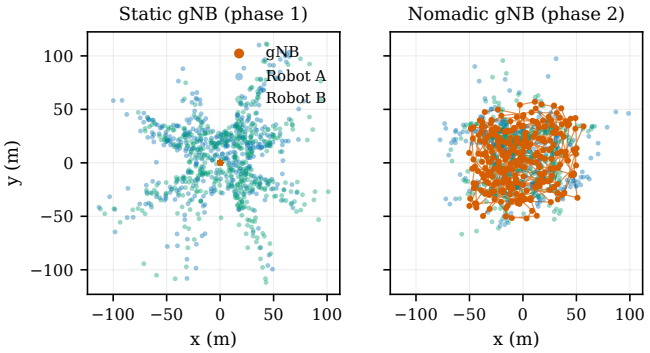


Fig. 2. Top-down gNB and robot positions in the urban world. In phase 1 (left) the gNB is fixed at the origin and robots roam widely; in phase 2 (right) the gNB random-walks and the coverage controller keeps robots clustered around its live ground track.

C2 availability as the fraction of samples with less than 100% C2 loss. A background watchdog logs every RLF recovery and every controller intervention.

Data hygiene. A single physical stack must serve only one trial: two concurrent trials share the same robots and gNB and silently corrupt each other, which we guard against with an exclusive campaign lock. Runs that predate the NLoS-aware controller, or that suffered the re-attach cascade, are excluded; the four reported worlds are clean 6-hour trials (733, 721, 482, and 974 samples for open field, urban, forest, and hills respectively).

VII. Results

A. C2 availability: the mobility effect flips with terrain

Figure 3 and Table I give the headline. In *open field*, gNB mobility *helps*: the weak-link robot's C2 availability rises from 56.5% (static) to 88.7% (nomadic), a +32 pp gain, because the wandering gNB periodically drifts closer and rescues a robot stranded near the coverage edge (its mean distance falls from 91 to 75 m). The strong robot is already near 98% in both phases. In *urban*, mobility is *neutral* (∼96–98% in both phases): localized buildings leave LoS pockets, and the controller keeps robots in them regardless of gNB motion. In *hills* and *forest*, mobility *hurts*, and the magnitude tracks how pervasive the blockage is: knife-edge terrain costs −10.6 pp (robot-averaged) while everywhere-present foliage costs −22 pp, collapsing both robots to ∼73–75%. Figure 4 summarizes the per-robot deltas.

B. Path loss and throughput explain the effect

Figure 5 bins measured path loss against gNB–robot slant distance per world. Open field and hills sit low (steep LoS-dominated curves); forest rises sharply because a longer slant path cuts through more canopy; urban is intermediate with building-driven jumps. Mobility moves robots along these curves: in forest, nomadism raises mean path loss from ∼11 to ∼22 dB and drives served TCP throughput from 15.1 to 3.2 Mb/s while fixed-rate UDP loss climbs above 70% (Fig. 6). In urban and open field, throughput is essentially preserved

TABLE I
C2 AVAILABILITY AND MEAN LINK METRICS BY WORLD AND GNB PHASE (ROBOT-AVERAGED UNLESS NOTED). Δ IS NOMADIC−STATIC C2.

World	Phase	C2 (%)	PLoss (dB)	SINR (dB)	TCP (Mb/s)
Open field	Static	76.8	16.5	35.1	9.7
	Nomadic	93.5	14.2	38.8	13.1
	Δ	**+16.7**			
Urban	Static	97.0	8.8	43.8	17.6
	Nomadic	96.9	10.4	42.6	16.1
	Δ	**−0.1**			
Hills	Static	98.9	3.9	45.7	21.6
	Nomadic	88.3	9.3	44.2	19.4
	Δ	**−10.6**			
Forest	Static	96.2	10.8	40.9	15.1
	Nomadic	74.2	21.7	32.5	3.2
	Δ	**−22.0**			

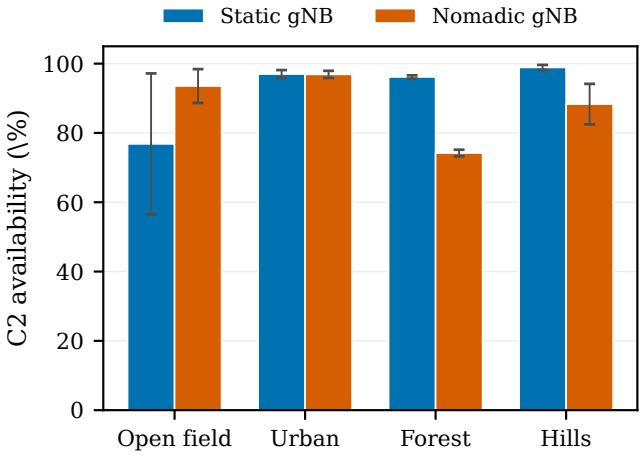


Fig. 3. C2 availability by world and gNB phase (bar = robot mean; whiskers span the two robots). Nomadism helps in open field, is neutral in urban, and hurts in hills and forest.

or improved. Figure 7 shows the SS-SINR distributions: all worlds are healthy in the LoS-dominated regime, but forest develops a heavy low-SINR tail under nomadism.

C. Link recovery: prevention beats reaction

Figure 8 counts link events per 6-hour run. With the NLoS-aware controller, the re-attach cascade is eliminated in urban and hills (zero failed re-attaches) and contained in forest (16), where the controller intervenes 574 times – roughly once every 37 s – to keep robots out of foliage dead zones. The cost of prevention is visible in Fig. 2: the controller keeps NLoS robots closer to the gNB than free roaming would, so their trajectories are more gNB-centred. This is the price of any valid measurement in NLoS: without it, the terrains do not merely degrade, they fail to produce data.

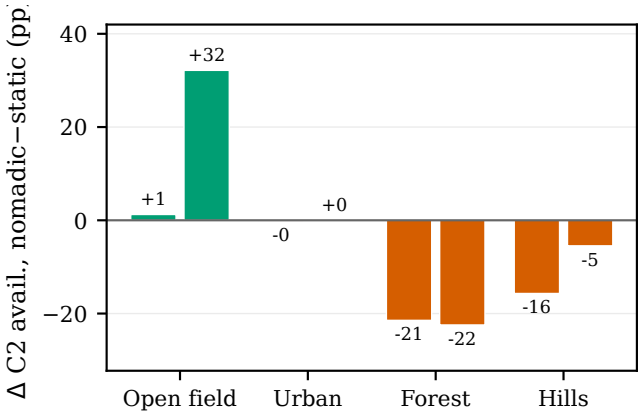


Fig. 4. Per-robot change in C2 availability from static to nomadic gNB. The weak-link robot in open field gains most; both robots lose most in forest.

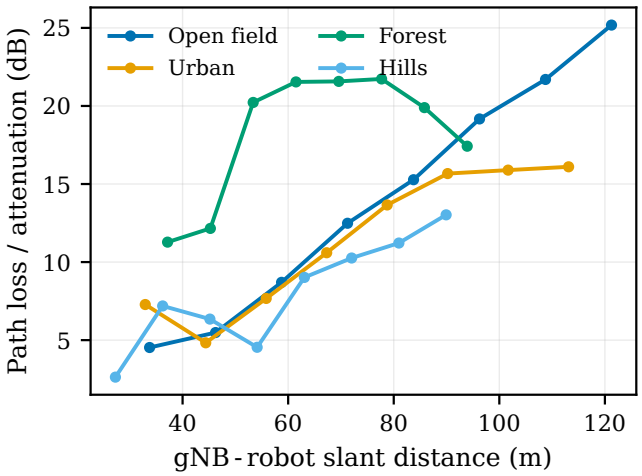


Fig. 5. Measured path loss vs. gNB–robot slant distance (binned means). Forest rises steeply with distance (foliage path length); hills and open field stay low.

D. Indoor: a coverage limit, not a tuning problem

The outdoor-to-indoor world is qualitatively different. The fixed wall-penetration term plus the link baseline places the *entire* building interior near the RLF cliff – depth barely matters – so shadow spikes trip RLF repeatedly, and each ∼2-minute re-attach dominates the timeline. Lowering the gNB to hover directly overhead (cutting ∼7 dB of A2G loss) and adding the retreat-to-door recovery removes the cascade and keeps robots alive, but C2 availability remains only 12–18% (Fig. 9). We interpret this as a genuine finding rather than a defect: a ground robot cannot hold C2 deep inside a building at this link budget, and *leaving the building is the only recovery* – exactly the behaviour the retreat controller encodes.

VIII. DISCUSSION

The unifying explanation is that gNB mobility changes the *propagation path*, and whether that helps depends on how blockage is distributed. In LoS, a moving gNB is pure

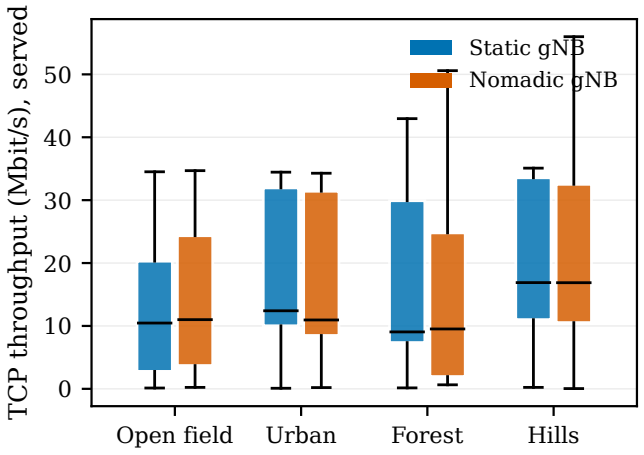


Fig. 6. Served TCP throughput by world and phase. Forest collapses under a nomadic gNB; urban and open field are preserved.

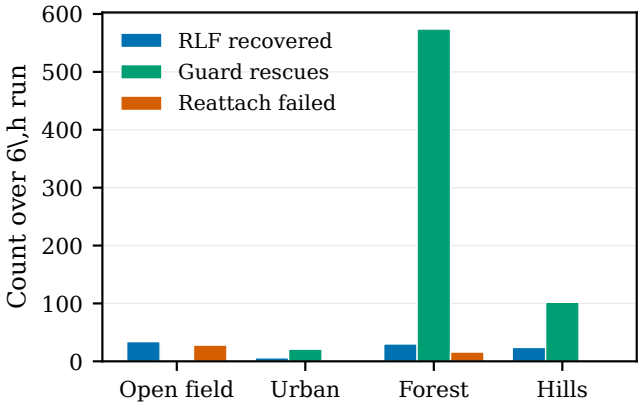


Fig. 8. Link events over each 6-hour run. Forest requires by far the most controller interventions; urban and hills incur zero failed re-attaches.

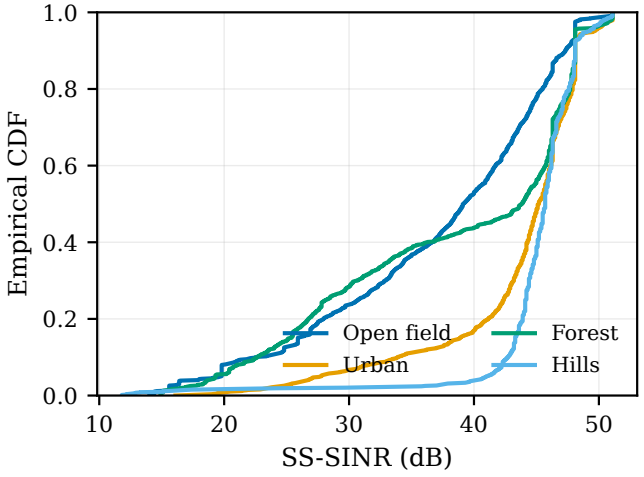


Fig. 7. Empirical CDF of SS-SINR. Forest develops a heavy low-SINR tail; other worlds remain in the high-SINR regime.

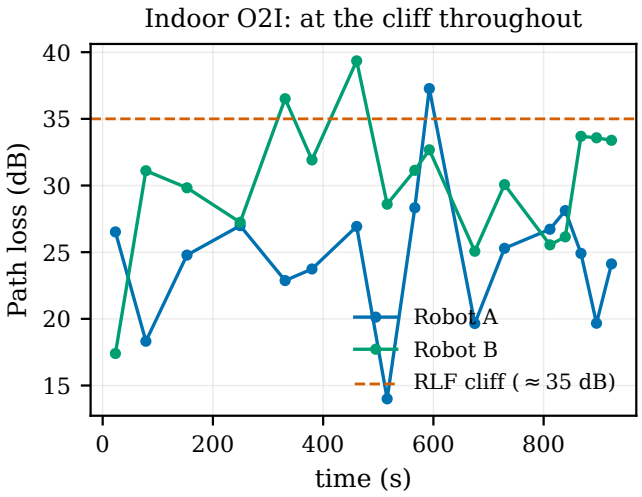


Fig. 9. Indoor O2I path loss over time (low-hovering gNB). Both robots sit at the RLF cliff throughout; the interior offers no reliably covered pocket.

geometric benefit: it can shorten links and rescue edge robots, so nomadism helps the weak robot most. Where blockage is *localized* (urban buildings), a controller can keep robots in LoS pockets and mobility washes out. Where blockage is *pervasive* (forest canopy), there is no pocket to hold: any repositioning that lengthens the ground path lengthens the foliage path, so mobility is strictly harmful, and the damage scales with blockage density (hills $<$ forest). For SAR planning this suggests a simple heuristic: *hover* a nomadic gNB when the environment is cluttered and pervasive, and *roam* it only when links are LoS-dominated or robots risk stranding at the coverage edge.

IX. Threats to Validity

Emulated channel. Attenuation is applied by the RF simulator from our A2G model; absolute values depend on model constants and the simulator's noise floor, so we emphasize *relative* static-vs-nomadic effects within a fixed calibration rather than absolute link budgets. **Controller coupling.** The coverage controller is necessary to obtain data in NLoS but alters robot trajectories (more gNB-centred), so the reported effects are conditional on prevention-based coverage keeping; free-roaming robots would simply lose the link. **Single cell, two UEs.** We study one gNB and two robots; multi-cell handover and many-robot contention are future work. **Sampling rate.** Under outage the expensive per-sample probes time out, so heavily degraded phases yield fewer samples (e.g. forest phase 2); fine-grained path loss is nonetheless logged continuously by the bridge. **Indoor calibration.** The O2I result reflects one building geometry and link budget; a different penetration term or baseline would move the coverage limit, though not the qualitative conclusion that deep O2I is the hardest case.

X. Conclusion

We built a hardware-in-the-loop emulation that runs a complete 5G SA stack against physics-simulated robot and UAV motion through a geometry-consistent air-to-ground channel, with namespace isolation that makes the command-and-control downlink physically honest. Using it to compare a static and a nomadic aerial gNB across four terrains for six hours each, we showed that the effect of gNB mobility on C2 availability is terrain-dependent and changes sign with the pervasiveness of blockage: helpful in line-of-sight (open field, $+32$ pp for the weak robot), neutral under localized urban NLoS, and increasingly harmful under diffraction and foliage (hills -10.6, forest -22 pp). We further showed that maintaining connectivity in NLoS at all requires a coverage controller that reacts to measured path loss rather than geometric distance, and that deep outdoor-to-indoor penetration is effectively uncoverable at this link budget, with retreat-to-door as the only recovery. The emulation stack, experiment automation, and dataset are released to support reproduction and extension to multi-cell, many-robot SAR scenarios.

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